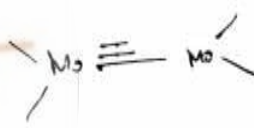


Part A

- The size of the d orbitals in Si, P, S and Cl follows the order—
 A) $Cl > S > P > Si$
 B) $Cl > P > S > Si$
 C) $P > S > Si > Cl$
 D) $Si > P > S > Cl$
- The oxidation state of iron in met-hemoglobin is—
 A) Three
 B) Two
 C) Four
 D) Zero
- As a ligand Cl^- is—
 A) Only a σ -donor
 B) Only a π -donor
 C) Both a σ -donor and a π -donor
 D) a σ -donor and a π -acceptor
- The number of metal-metal bond(s), with σ , π , and δ character, present in $[Mo_2(CH_3CO_2)_4]$ complex is(are), respectively,
 A) 1, 2, 1
 B) 1, 2, 0
 C) 1, 1, 0
 D) 1, 1, 1


- The least basic among the following is —
 A) $Al(OH)_3$
 B) $La(OH)_3$
 C) $Ce(OH)_3$
 D) $Lu(OH)_3$
- The correct d-electron configuration showing spin-orbit coupling is—
 A) $t_{2g}^6 e_g^2$
 B) $t_{2g}^6 e_g^0$
 C) $t_{2g}^4 e_g^0$
 D) $t_{2g}^3 e_g^2$
- Among the following, the strongest oxidizing agent is—
 A) $[WO_4]^{2-}$
 B) $[CrO_4]^{2-}$
 C) $[MoO_4]^{2-}$
 D) $[ReO_4]^{-1}$



8. The number of Ni-Ni bonds in $[\text{CpNi}(\mu\text{-PPh}_2)]_2$ complex obeying the 18 electron rule is
 A) 0
 B) 1
 C) 2
 D) 3
9. The number of microstates for d^5 electron configuration is—
 A) 21×6^3
 B) 14×6^3
 C) 7×6^2
 D) 28×6^3
10. Consider σ -bonding only, in the MO diagram of a metal complex with trigonal bipyramidal (TBP) geometry, the d orbitals which remain non-bonding are :
 A) dz^2 and dxz
 B) dxz and dyz
 C) dx^2-y^2 and dxy
 D) dz^2 and dyz
11. The reaction of IO_3^- with I^- in aqueous acidic medium results in
 A) I_2 and H_2O
 B) I_2 and H_2O_2
 C) IO^- and H_2O
 D) IO^- and H_2O_2
12. The metal complex that exhibits a triplet as well as a doublet in its ^{31}P NMR spectrum is—
 A) $\text{mer-}[\text{IrCl}_3(\text{PPh}_3)_3]$
 B) $\text{trans-}[\text{IrCl}(\text{CO})(\text{PPh}_3)_2]$
 C) $\text{fac-}[\text{IrCl}_3(\text{PPh}_3)_3]$
 D) $[\text{Ir}(\text{PPh}_3)_4]^+$
13. Structure of a carborane with formula, $\text{C}_2\text{B}_4\text{H}_8$ is formally derived from—
 A) Closo-borane
 B) Nido-borane
 C) Arachno-borane
 D) Conjuncto-borane
14. The number of metal-metal bonds in $\text{Ir}_4(\text{CO})_{12}$ is—
 A) 4
 B) 6
 C) 10
 D) 12
15. The reaction of decaborane $\text{B}_{10}\text{H}_{14}$ with acetylene in the presence of Et_2S gives
 A) $\text{C}_2\text{B}_{10}\text{H}_{12}$
 B) $\text{C}_2\text{B}_8\text{H}_{10}$
 C) $\text{C}_2\text{B}_{10}\text{H}_{14}$
 D) $\text{C}_2\text{B}_9\text{H}_{11}$

- A) tricapped octahedron
B) monocapped cube
C) tricapped trigonal prism
D) heptagonal bipyramid
- 24.
17. The cluster type and geometry of the species $[\text{Rh}_9\text{P}(\text{CO})_{21}]^{2-}$ are
A) closo, tricapped trigonal prism
B) arachno, trigonal prism
C) nido, capped square antiprism
D) nido, bicapped trigonal prism
- 25.
18. In compound $\text{N}_3\text{P}_3\text{F}_6$, the geometry around nitrogen and phosphorus, respectively, are:
A) pyramidal and tetrahedral
B) planar and tetrahedral
C) pyramidal and planar
D) planar and trigonal bipyramidal
- 26.
19. The standard reduction potentials in acid medium for F_2 , Cl_2 , Na and Zn are in the order:
A) $\text{F}_2 > \text{Cl}_2 > \text{Na} > \text{Zn}$
B) $\text{F}_2 > \text{Cl}_2 > \text{Zn} > \text{Na}$
C) $\text{Na} > \text{Zn} > \text{Cl}_2 > \text{F}_2$
D) $\text{Cl}_2 > \text{F}_2 > \text{Zn} > \text{Na}$
- 27.
20. Which of the following is amphoteric?
A) CrO
B) Cr_2O_3
C) CrO_5
D) CrO_3
- 28.
21. Which compound is used in ultra-violet calibration?
A) Hg_2Cl_2
B) HgCl_2
C) $\text{K}_2\text{Cr}_2\text{O}_7$
D) KMnO_4
22. In lanthanides, Ce changes from
A) +4 to +3 oxidation state
B) +1 to +3 oxidation state
C) +2 to +3 oxidation state
D) +3 to +2 oxidation state
23. The cluster having arachno type structure is—
A) $[\text{Os}_5(\text{CO})_{16}]$
B) $[\text{Os}_3(\text{CO})_{12}]$
C) $[\text{Ir}_4(\text{CO})_{12}]$
D) $[\text{Rh}_6(\text{CO})_{16}]$
- 29.

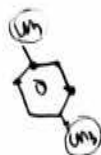
24. The number of spin-allowed ligand field transitions for octahedral Ni(II) complexes with $^3A_{2g}$ ground state is—
 A) Two
 B) Three
 C) One
 D) Four
25. The final product of the reaction $[Mn(CO)_6]^+ + MeLi \rightarrow$ is—
 A) $[Mn(CO)_6]^- Me$
 B) $[Mn(CO)_5 Me]$
 C) $[Mn(CO)_6]$
 D) $[(MeCO)Mn(CO)_5]$
26. In photosynthetic systems the redox metalloproteins involved in electron transfer are cytochrome (cyt b), cytochrome bf complex (cyt bf) and plastocyanin (PC). The pathway of electron flow is—
 A) $PC \rightarrow cyt\ b \rightarrow cyt\ bf$
 B) $cyt\ bf \rightarrow cyt\ b \rightarrow PC$
 C) $cyt\ b \rightarrow cyt\ bf \rightarrow PC$
 D) $PC \rightarrow cyt\ bf \rightarrow cyt\ b$
27. The correct set of the biologically essential elements is—
 A) Fe, Mo, Cu, Zn ✓
 B) Fe, Cu, Co, Ru
 C) Cu, Mn, Zn, Ag ✗
 D) Fe, Ru, Zn, Mg
28. The platinum complex of NH_3 and Cl^- ligands is an anti-tumour agent. The correct isomeric formula of the complex and its precursor are—
 A) $cis-Pt(NH_3)_2Cl_2$ and $PtCl_4^{2-}$
 B) $trans-Pt(NH_3)_2Cl_2$ and $PtCl_4^{2-}$
 C) $cis-Pt(NH_3)_2Cl_2$ and $Pt(NH_3)_4^{2+}$
 D) $trans-Pt(NH_3)_2Cl_2$ and $Pt(NH_3)_4^{2+}$
29. Based on the behaviour of the metalloenzymes, consider the following statements :
 1. In the enzymes, the zinc activates O_2 to form peroxide species.
 2. In the enzymes, the zinc activates H_2O and provides a zinc bound hydroxide.
 3. In the oxidases, the iron activates O_2 to break the bonding between the two oxygens.
 4. Zinc ion acts as a nucleophile and attacks at the peptide carbonyl.
 The set of correct statements is—
 A) 1 and 2
 B) 2 and 3
 C) 3 and 4
 D) 1 and 4

30. Which of the following prevents the inactivation of nitrogenase by oxygen?
 A) Cytochrome
 B) Carotene
 C) Xanthophyll
 D) Leghaemoglobin
31. A neutron beam falls from a crystal of interplanar spacing $3.84 \text{ \AA}$ with an angle of reflection 30° ; the speed of the neutron will be
 A) $1.03 \times 10^4 \text{ cm/sec}$
 B) $1.03 \times 10^3 \text{ cm/sec}$
 C) $1.03 \times 10^5 \text{ cm/sec}$
 D) $1.03 \times 10^6 \text{ cm/sec}$
32. An element with a BCC structure has an edge length of 288 pm and a density of 7.2 g cm^{-3} . How many atoms of the element are there in 208 g ?
 A) 24.16×10^{26}
 B) 24.16×10^{24}
 C) 24.16×10^{23}
 D) 24.16×10^{25}
33. If the edge length of a BCC crystal of an element is $A \text{ cm}$, M is the atomic mass and N_0 is the Avogadro number, the density of the crystal will be
 A) $2N_0/A^3M$
 B) $4M/A^3N_0$
 C) $A^3M/2N_0$
 D) $2M/A^3N_0$
34. In polarography, if 'M' is the mass of the mercury drop and 'T' is the drop time, the diffusion current is proportional to
 A) $M^{2/3}T^{1/6}$
 B) $M^{3/2}T^{1/6}$
 C) $M^{2/3}T^{1/3}$
 D) $M^{3/2}T^{1/3}$
35. Choose the correct statement(s)
 (i) The obtained polarogram for each electroactive species exhibits a sigmoidal curve.
 (ii) In Coulometry, the analyte is either partially oxidized or reduced.
 (iii) Half-wave potentials are dependent on the concentration of electroactive species.
 (iv) The residual faradaic current that arises from O_2 as impurity is eliminated by purging the analyte solution with N_2 .
 A) i and iii
 B) i, ii and iv
 C) ii, iii and iv
 D) i and iv
36. Two solutes to attempt to
 A) Use
 B) Use
 C) Use
 D) Use
37. The $^1\text{H-NMR}$
 A) Ten
 B) Two
 C) Two
 D) Two
38. The carbon
 A) 16
 B) 17
 C) 17
 D) 17
39. The correct
 A)
 B)
 C)
 D)
40. The
- 41.

36. Two solutes with short retention times co-elute in gas chromatography; what is the simplest way to attempt to resolve the peaks
- Use a higher column temperature
 - Use a more polar solvent
 - Use a longer column
 - Use a lower column temperature

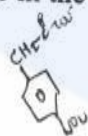
37. The ^1H -NMR spectrum of 1,4-dimethylbenzene will have

- Ten singlets
- Two singlets
- Two doublets and one singlet
- Two doublets and two singlets



38. The carbonyl absorption band of *p*-nitrophenylacetate in the IR spectrum appears at

- 1670 cm^{-1}
- 1700 cm^{-1}
- 1730 cm^{-1}
- 1760 cm^{-1}



39. The correct order of the NMR chemical shift (δ) of the following compounds is



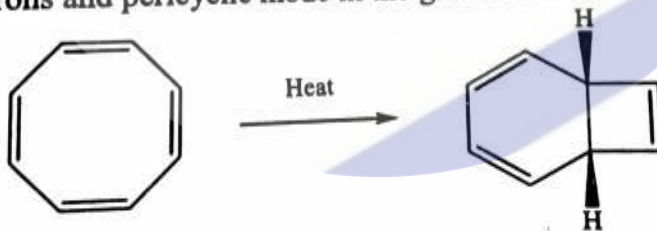
I

II

III

- I: 5.4; II: 9.2; III: 7.2
- I: 9.2; II: 5.4; III: 7.2
- I: 5.4; II: 7.2; III: 9.2
- I: 9.2; II: 7.2; III: 5.4

40. The number of π electrons and pericyclic mode in the given reaction are

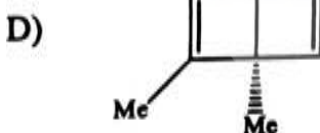
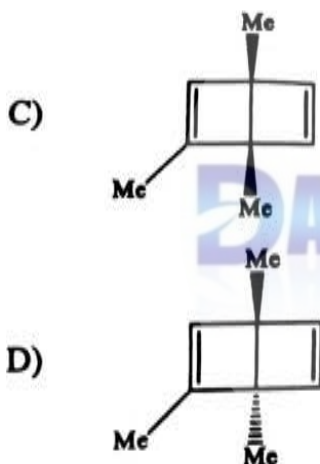
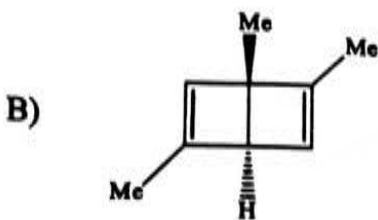
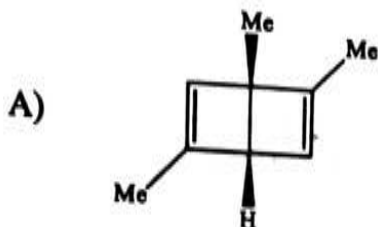
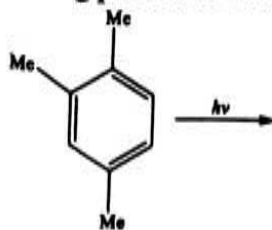


- 6 and Conrotatory
- 6 and disrotatory
- 4 and Conrotatory
- 4 and disrotatory

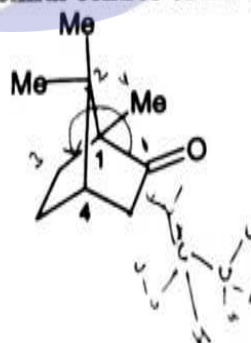
41. Which of the following diene is more reactive in the Diels-Alder reaction

- (4E)-1,4-hexadiene
- (2E, 4E)-2,4-hexadiene
- (4Z)-1,4-hexadiene
- (2Z, 4Z)-2,4-hexadiene

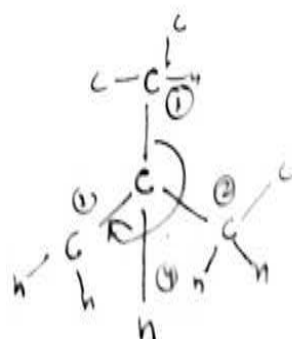
42. The major product formed in the following photochemical reaction is



43. The absolute configuration of the two chiral centres of the following compound is



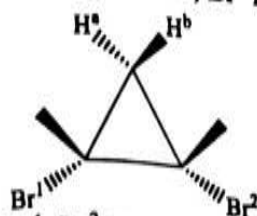
- A) 1R, 4S
B) 1R, 4R
C) 1S, 4R
D) 1S, 4S



44. The biosynthetic precursor of the steroids is

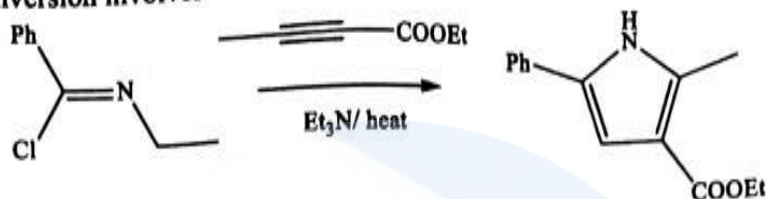
- A) Shikmic acid
B) Mevalonic acid
C) Secologanin
D) α -ketoglutaric acid

45. What is the relation between H^a , H^b , and between Br^1 , Br^2 in the given compound



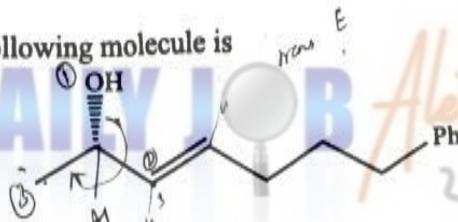
- A) H^a , H^b are diastereotopic; and Br^1 , Br^2 are enantiotopic
 B) H^a , H^b are enantiotopic; and Br^1 , Br^2 are diastereotopic
 C) H^a , H^b are enantiotopic; and Br^1 , Br^2 are homotopic
 D) H^a , H^b are diastereotopic; and Br^1 , Br^2 are homotopic

46. The reaction conversion involves



- A) a 1,3-dipolar species as reactive intermediates
 B) a carbenium ion as reactive intermediate, and a cycloaddition
 C) a 1,3-dipolar species as reactive intermediate, and an aza witting reaction
 D) a carbanion as reactive intermediate, and an aza Cope rearrangement

47. The IUPAC name for the following molecule is

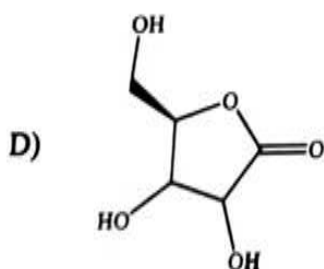
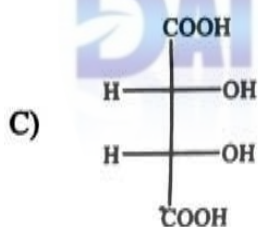
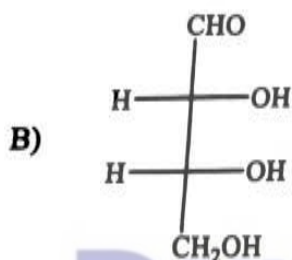
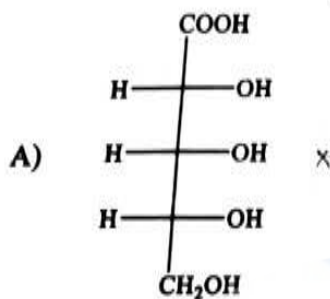
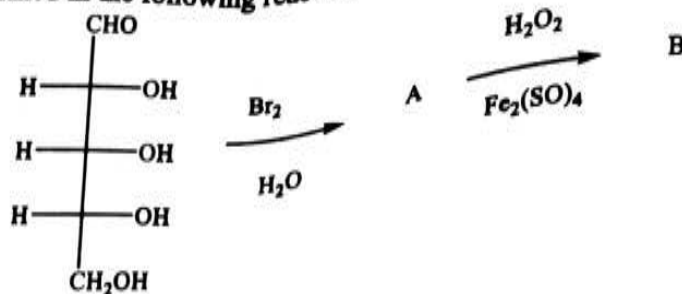


- A) (2R,3Z)-7-phenylhept-3-en-2-ol
 B) (2S,3Z)-7-phenylhept-3-en-2-ol
 C) (2R,3E)-7-phenylhept-3-en-2-ol
 D) (2S,3E)-7-phenylhept-3-en-2-ol

48. The heterocyclic ring present in the amino acid histidine is

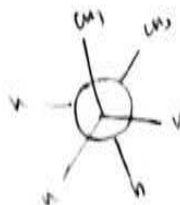
- A) Indole
 B) Pyridine
 C) Imidazole
 D) Tetrahydropyrrolo

49. Major product B formed in the following reaction is

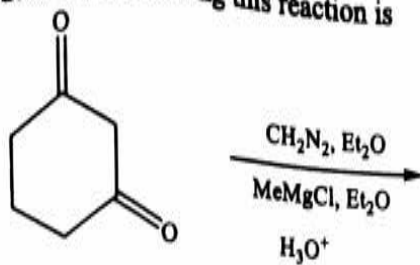


50. The gauche confirmation of butane at $\theta = 60^\circ$ possesses

- A) Plane of symmetry and achiral
- B) C_2 -axis of symmetry and chiral
- C) Centre of symmetry and achiral
- D) Plane of symmetry and chiral

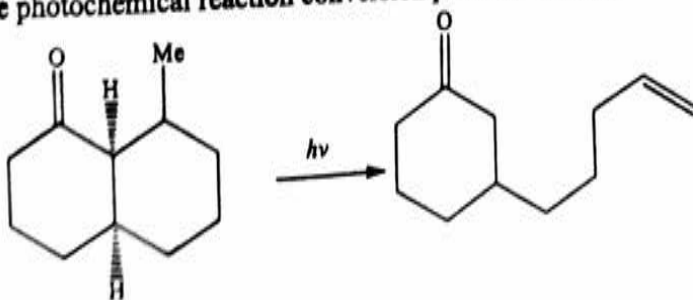


51. The structure of the major product formed during this reaction is



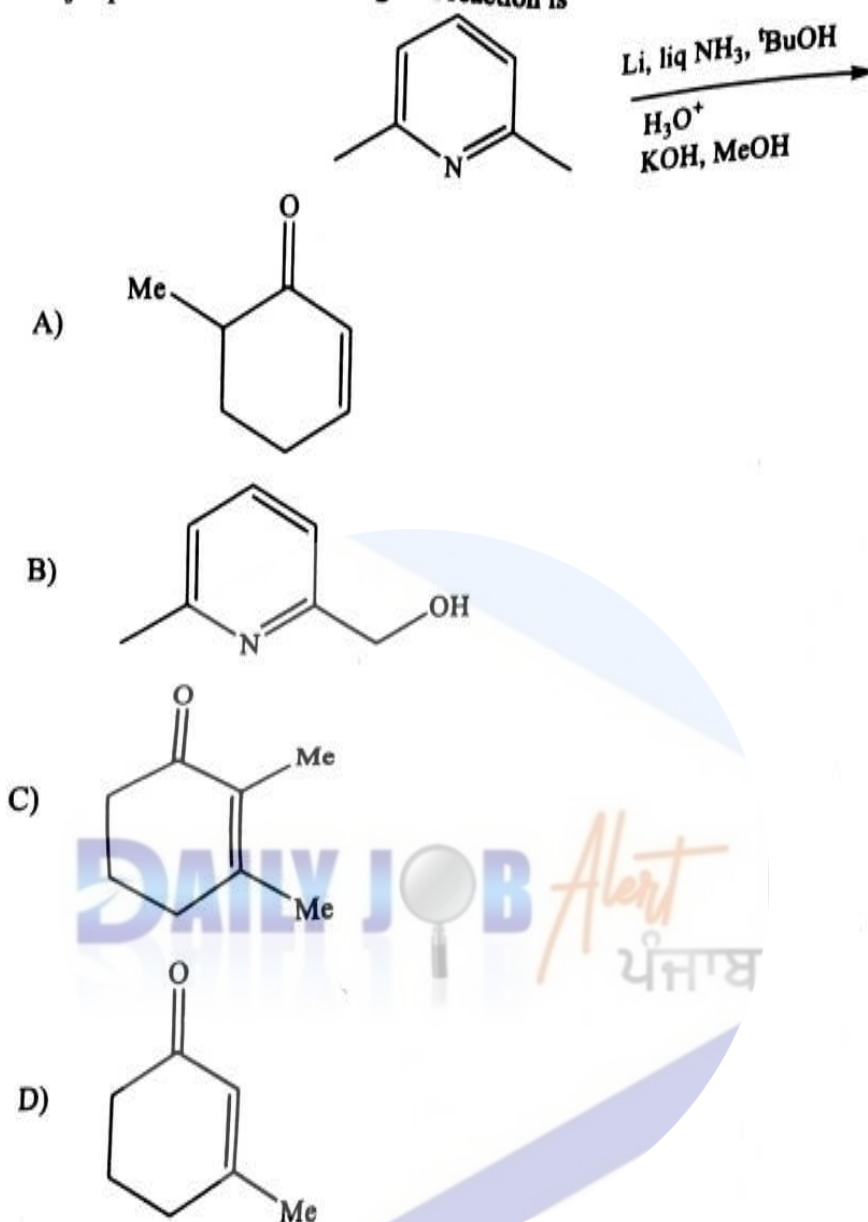
- A)
- B)
- C)
- D)

52. The photochemical reaction conversion proceeds through



- A) Norrish type-I reaction
 B) Norrish type-II reaction
 C) Barton reaction
 D) Paterno-Buchi reaction

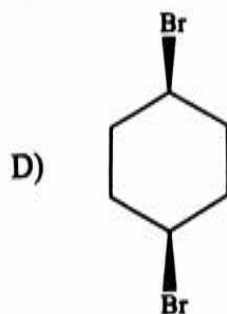
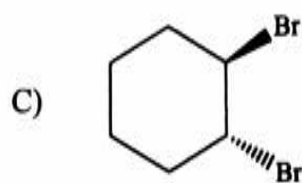
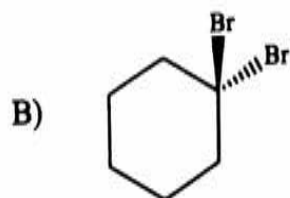
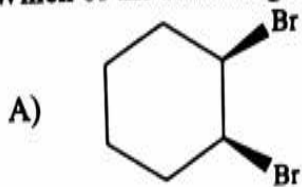
53. Major product formed in the given reaction is



54. In which reaction condition, nucleophilic substitution reaction of pyridine is most effective

- A) In slightly basic conditions
- B) In slightly acidic conditions
- C) In a neutral medium
- D) No specific condition requirements

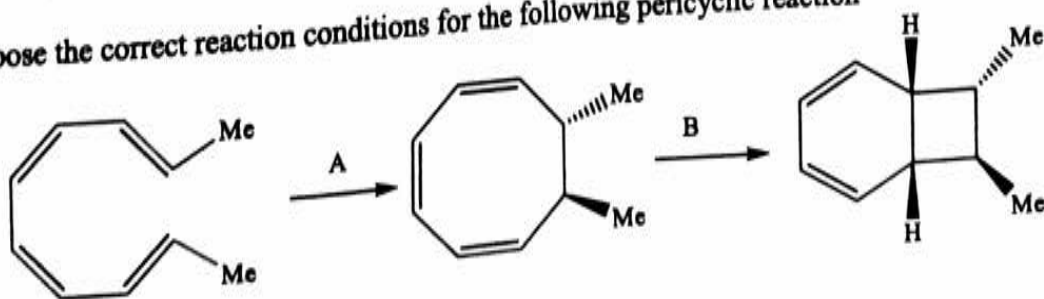
55. Which of the following dibromocyclohexane, reacts faster with NaI to give cyclohexane



56. The source emitting at 14.4 Kev (3.48×10^{18} Hz) has to move towards the absorber at 2.2 mm s^{-1} for resonance in the Mossbauer experiment. The shift in the frequency between the source and the absorber is

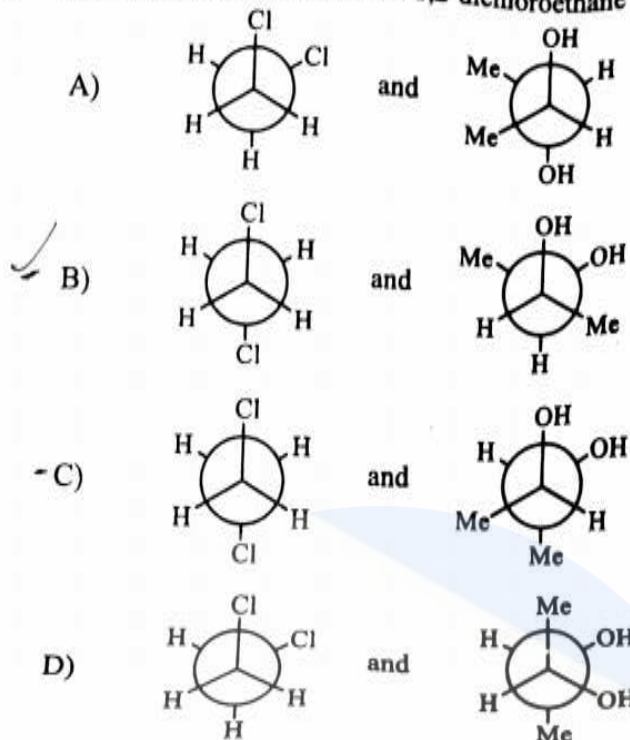
- A) 15.0 MHz
B) 20.0 MHz
C) 25.5 MHz
D) 30.0 MHz

57. Choose the correct reaction conditions for the following pericyclic reaction

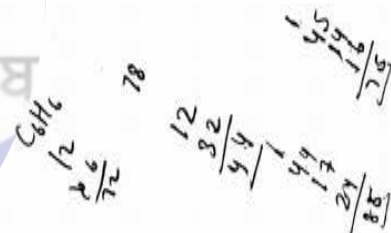
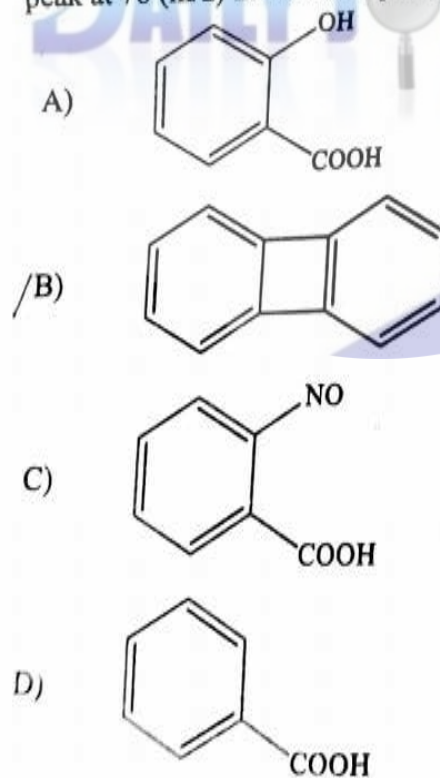


- A) A- Δ ; B- Δ
B) A- $h\nu$; B- Δ
C) A- $h\nu$; B- $h\nu$
D) A- Δ ; B- $h\nu$

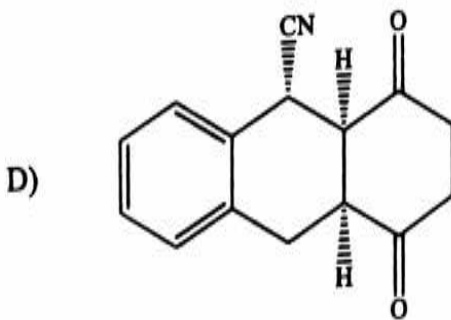
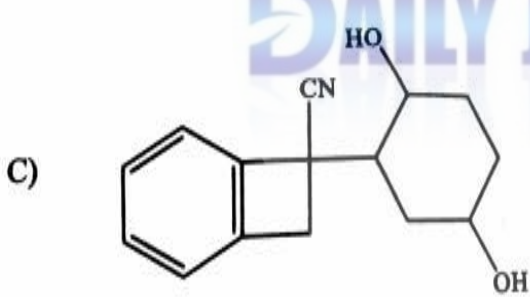
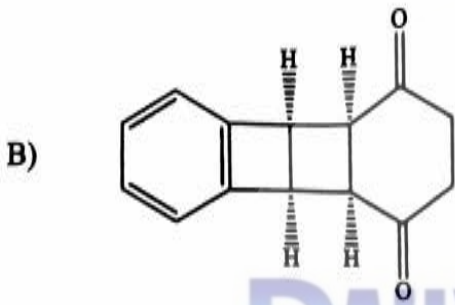
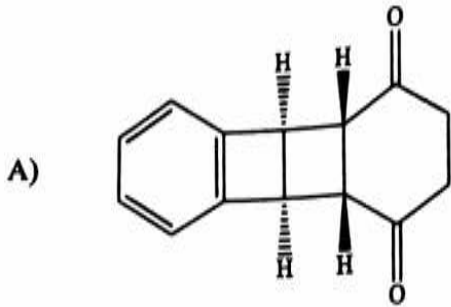
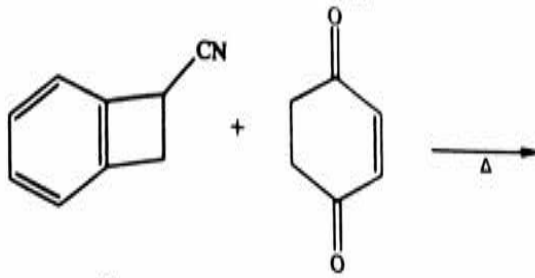
58. Most stable conformations of 1,2-dichloroethane and *dl*-2,3-butanediol are



59. The reaction of anthranilic acid with *iso*-amyl nitrite gave a product which displays a strong peak at 76 (m/z) in its mass spectrum, which would be



60. The major product formed from the following reaction is



61. The wave function of a particle in a box is given by $\psi = A \sin n\pi \frac{x}{L}$, where $A =$
- $\sqrt{\frac{2}{L}}$
 - $\sqrt{\frac{2L}{h}}$
 - $\sqrt{\frac{L}{2}}$
 - $\sqrt{\frac{2h}{L}}$
62. The probability of finding the particle in a box of width L , at $x=L$ is
- 1
 - 0
 - 0.5
 - 0.75
63. How many NMR signals would be given by the compound $(CH_3)_2CHCH_2CH_3$
- 1
 - 2
 - 3
 - 4
64. The distance between the centers of the peaks of doublet is called as?
- Coupling constant
 - Spin constant
 - Spin-spin coupling
 - Chemical shift
65. Full form of NMR is
- Nuclear Magnetic Reaction
 - Nuclear Magnetic Resonance
 - Nuclear Mass Reaction
 - Nuclear Magnetic Resonances
66. Phase rule is applicable for ----- systems with equilibrium.
- Homogeneous
 - Heterogeneous
 - Reversible
 - Irreversible
67. The mathematical expression for Phase rule is
- $F = C - P + 2$
 - $F = C + P + 2$
 - $P = F - P + 2$
 - $C = P - F + 2$

68. What is the correct expression of $k(T)$ in terms of fugacity?
- $\prod_{i=1}^N \left(\frac{f_i}{f_i^0}\right)^{\nu_i}$
 - $\prod_{i=1}^N \left(\frac{f_i^0}{f_i}\right)^{\nu_i}$
 - $\sum_{i=1}^N \left(\frac{f_i}{f_i^0}\right)^{\nu_i}$
 - $\sum_{i=1}^N \left(\frac{f_i^0}{f_i}\right)^{\nu_i}$
69. As time progresses, Energy will naturally become more spread out. It can only come together if you make it.
- Zeroth Law of Thermodynamics
 - First Law of Thermodynamics
 - Second Law of Thermodynamics
 - Third Law of Thermodynamics
70. For the reaction, $\text{SO}_2 + \text{O}_2 \leftrightarrow \text{SO}_3$
 If the concentration of SO_2 is increased, the equilibrium of the reaction will shift _____.
- Left
 - Right
 - Left and right
 - Neither left nor right
71. The single-particle partition function (f) for a certain system has the form $f = AVe^{BT}$. The average energy per particle will then be (K = Boltzmann constant)
- BKT
 - BKT^2
 - KT/B
 - KT/B^2
72. The pressure of an Ideal gas in a vessel is p . The gas is heated at constant volume until the r.m.s speed of the gas molecules is doubled. What is the new pressure of the gas?
- $\sqrt{2}p$
 - $2p$
 - $3p$
 - $4p$
73. Suppose a box contains about 5.0×10^{21} hydrogen atoms at room temperature (21°C). Determine the thermal energy of these atoms.
- 10 J
 - 20 J
 - 30 J
 - 5 J

248

$$\frac{23}{49}$$
$$\begin{array}{r} 83 \\ 69 \overline{) 5727} \\ \underline{508} \\ 647 \\ \underline{572} \\ 77 \end{array}$$

75. Calculate the temperature for which K_{eq} for a reaction is 1.04×10^3 where $\Delta H^\circ = -83.2 \text{ kJ/mol}$ and $\Delta S^\circ = -246 \text{ J/mol} \cdot \text{K}$.
- A) 0.307 K
 B) 307 K
 C) 274 K
 D) 0.274 K

DG = RT ln K

2.2.303 RT 19 X
2.303 X 6.34 X

2-2302
2-2303X
2-2303X8-3149

$$-2.303 \times 6.34 \times 37 = -83200 + 2467$$

10. c

- $$T = \frac{83200}{303.27} K$$

- $$\frac{7283200}{303,27}$$

- $$\begin{array}{r} 203 \overline{) 83200} \\ \underline{606} \\ 22600 \end{array}$$

- $$\begin{array}{r} 2 \\ 8320 \\ 606 \end{array}$$

81. What is the energy needed for reactant particles to change into products?
- Potential
 - Kinetic
 - Activation
 - Conversion
82. What would be the overall order of the rate law, $\text{Rate} = [A]^2[B]^1$
- Zero order
 - 1st order
 - 2nd order
 - 3rd order
83. The eigen value obtained when the operator $\frac{d^2}{dx^2}$ operates on the eigen function 'sin x' is
- cos x
 - 0
 - +1
 - 1
84. What is explaining behaviours of a system at equilibrium?
- Avogadro's Law
 - Boyle's Law
 - Le-Chatelier's Principle
 - Collision Theory
85. In a homogeneous catalysis
- The catalyst and the reactants should be gases
 - The catalyst and the reactants should be from a single phase
 - Catalyst and the reactants are all solids
 - The catalyst and the reactants are all liquids
86. Colloidal system is-----
- Homogeneous
 - Heterogeneous and translucent
 - Heterogeneous and transparent
 - Homogeneous and translucent
87. The decrease in the conductance of the solution when NaOH is added to HCl is due to
- Decrease in the concentration
 - Decrease in the temperature
 - Decrease in the mobility
 - None of the above

88. Colloidal particles carry the charge....

- A) Negative
- B) Positive
- C) Positive or negative
- D) No charge

89. The extent of adsorption of a gas on a solid depends on _____.

- A) Temperature of gas
- B) Pressure of gas
- C) Nature of the gas
- D) All are correct

90. A plot of $\log x/m$ versus $\log p$ for the adsorption of a gas on a solid gives a straight line with slope equal to

- A) n
- B) $1/n$
- C) $\log K$
- D) $-\log K$

$$\frac{x}{m} = K P^{1/n}$$

$$\log \frac{x}{m} = \frac{1}{n} \log P + \log K$$

Part B

91. Which sport's competition is known as the "Korn Ferry Tour"?

- A) Golf
- B) Baseball
- C) Tennis
- D) Chess

92. The term "Boycott of British Goods" was used in which of the following movements?

- A) Swadeshi Movement
- B) Quit India Movement
- C) Non Cooperative Movement
- D) Home Rule Movement

93. Rani-ki-Vav (the Queen's Stepwell), which is in the UNESCO World Heritage List, is located in which state?

- A) Gujarat
- B) Madhya Pradesh
- C) Rajasthan
- D) Uttar Pradesh

94. Which gas is most effective for ozone depletion?

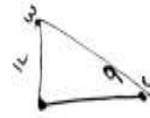
- A) CMC
- B) CFC
- C) CCF
- D) CDC

95. In which year was "The National Commission for Women" set up?
 A) 1992
 B) 1990
 C) 2015
 D) 2012
96. Who was honoured for film and television achievements at the MTV Movie & TV Awards 2022?
 A) Jennifer Lopez
 B) Tom Cruise
 C) Will Smith
 D) Angelina Jolie
97. What is the repo rate after August, 2022 Monetary Policy Committee meeting?
 A) 5.0%
 B) 5.2%
 C) 5.4%
 D) 5.75%
98. Which state government has partnered with Linked In to boost employability skills of its students?
 ✓ A) Punjab
 B) Haryana
 C) Kerala
 D) Tamilnadu
99. Who has written the book "New Dimensions of India's Foreign Policy"?
 A) Raghuram Rajan
 B) Amartaya Sen
 C) Atal Bihari Vajpayee
 D) Amit Chaudhari
100. Which global organisation launched a new initiative called COVID-19 Technology Access Pool (C-TAP)?
 A) World Health Organisation
 B) World Trade Organisation
 C) UNICEF
 D) World Bank
101. Answer the question based on following information.
 I. 'P x Q' means 'P is the brother of Q'.
 II. 'P+Q' means 'P is the father of Q'.
 III. 'P / Q' means 'P is the sister of Q'.
 Which of the following represents 'P is the uncle of Q'?
 A) $P + D / Q$
 B) $P \times D + Q$
 C) $P + D \times Q$
 D) $P / D + Q$

Q2 me
P = det
03/04/23

102. Ram went to his office, which is 9 kms towards the South from his hostel. His roommate Meghal went to a clinic, which is 12 kms towards the West from the hostel. What is the shortest distance between the clinic and Ram's office?

A) 12 m
B) 12 kms
C) 15 kms
D) 15 m



$$\begin{array}{r} 12 \\ 9 \\ \hline 15 \end{array}$$

103. In a certain language 'pic vic pic' means 'winter is cold'; 'to nic re' means 'summer is hot'; 're it pa' means 'nights are hot'; Which of the following is the code for 'summer'?

A) to
B) nic
C) pic
D) vic

104. Six years ago, the ratio of the ages of Kunal and Sagar was 6:5. Four years hence, the ratio of their ages will be 11:10. What is Sagar's age at present?

A) 15
B) 16
C) 17
D) 18

$$\frac{3}{4} \times \frac{18}{5} = \frac{54}{5} = 10.8$$

$$\begin{array}{lcl} & K & S \\ 6 & \rightarrow & 6 \\ 4 & \rightarrow & 11 : 10 \end{array}$$

105. A wholesale tea dealer has 408 kilograms, 468 kilograms and 516 kilograms of three different qualities of tea. He wants it all to be packed into boxes of equal size without mixing. Find the capacity of the largest possible box.

A) 50
B) 36
C) 24
D) 12

$$\begin{array}{l} 408 : 468 : 516 \\ 68 : 78 : 86 \\ 34 : 39 : 43 \end{array}$$

$$\begin{array}{r} 136 \\ 156 \\ 172 \\ \hline 464 \end{array}$$

106. A simple mechanical device has four gears A, B, C and D such that they mesh in the order A-B-C-D. The number of teeth in these gear is 10, 21, 12, 17 respectively. If we start the mechanism from rest, how many revolutions will the largest gear have to turn before one full cycle is completed and the gears are in the position from where they started?

A) 270
B) 340
C) 7140
D) 285

107. If the LCM of two numbers is 750 and their product is 18750. Find the HCF of the numbers.

A) 40
B) 35
C) 25
D) 45

$$6:5$$

$$15$$

$$\begin{array}{r} 40 \overline{) 18750} \\ 40 \\ \hline 5 \end{array}$$

$$11:10$$

108. An amount of 2430 is divided among A, B and C such that if their shares be reduced by Rs 5, Rs.10 and Rs.15 respectively, the remainders shall be in the ratio of 3:4:5. Then, B's share was:
- A) 605
B) 800
C) 790
D) 810
109. From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while travelling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same directions as the train. What is the length of the train and that of the platform in meters, respectively
- A) 210 and 140
B) 162.5 and 187.5
C) 245 and 130
D) 175 and 200
110. A contract is to be completed in 56 days and 104 men were set to work, each working 8 hours a day. After 30 days, $\frac{2}{5}$ of the work is finished. How many additional men may be employed so that work may be completed on time each man now working 9 hours per day?
- A) 46 men
B) 48 men
C) 52 men
D) 56 men
111. Change into indirect speech. She said, "Could you bring my book tonight?"
- A) She asked me could I bring her book that night.
B) She asked me to bring her book that night.
C) She had asked me to bring her book that night.
D) She asked me to bring her book tonight.
112. Playing the piano may seem *very difficult* at first, but it gets easier with practice. Replace the italicized word with the correct substitute:
- A) manageable
B) sparse
C) arduous
D) ominous
113. Choose the correct phrasal or prepositional verb for 'to enter by force':
- A) break into
B) break away
C) break down
D) break out

114. Choose one word substitute for an action that is likely to make people very angry.
- inflationary
 - inflammable
 - commensurable
 - inflammatory
115. Many older people feel a slight pang of ... as they think back on their schools days.
- anger
 - tears
 - Pride
 - nostalgia
116. ਅਹਿਕਰੁ ਕਰੇ ਸੁ ਅਹਿਕਰੁ ਪਾਏ ਅਖਾਣ ਦਾ ਅਰਥ ਕੀ ਹੈ?
- ਗਰੀਬ ਨਾਲ ਸਦਾ ਜੱਗੋਂ ਬਾਹਰੀ
 - ਅਮੀਰ ਦੀ ਸਦਾ ਧੱਕਾ-ਜ਼ੋਰੀ
 - ਅਭਿਮਾਨੀ ਦੀ ਸਦਾ ਪੁੱਗਣੀ
 - ਕਰਨੀ ਅਨੁਸਾਰ ਫਲ
117. ਜਿਸ ਕੁਝੀ ਵਿੱਚ ਮੁੰਡਿਆਂ ਵਾਲੀਆਂ ਆਦਤਾਂ ਹੋਣ ਉਸ ਨੂੰ ਕੀ ਆਖਦੇ ਹਨ?
- ਜਨਾਨੜਾ
 - ਮੁੰਛੂ
 - ਮਾਹੀ-ਮੁੰਡਾ
 - ਖੁਸਰਾ
118. Time Barred ਦਾ ਅਰਥ ਹੈ:
- ਮੁਅੱਤਲੀ ਸਮਾਂ
 - ਸੀਮਤ ਸਮਾਂ
 - ਮਿਆਦ ਪੁੱਗੀ
 - ਮਿਤੀ ਬੱਧ
119. ਹੇਠ ਲਿਖਿਆ ਵਿੱਚੋਂ ਸੁੱਧ ਸ਼ਬਦ ਚੁਣੋ।
- ਬੋਹਤ
 - ਸੋਣਾ
 - ਗ਼ਜ਼ਲ
 - ਦੋੜ
120. ਮੈਂ ਆਪ ਉਸ ਨੂੰ ਚਾਹ ਪਾਰਟੀ ਤੇ ਛੱਡਣ ਗਿਆ 'ਵਾਕ ਵਿਚ' ਆਪ 'ਦਾ ਪ੍ਰਕਾਰਜ ਹੈ।
- ਨਿੱਜ ਵਾਚਕ ਪੜਨਾਵ
 - ਪੁਰਖ ਵਾਚਕ ਪੜਨਾਵ
 - ਸਬੰਧਵਾਚੀ ਪੜਨਾਵ
 - ਨਿਸ਼ਚੇ ਵਾਚਕ ਪੜਨਾਵ